Guided Exercise: VPC Network Peering within the Same Project

Overview & Objectives

In this hands-on guided exercise, you will create two separate custom Virtual Private Cloud (VPC) networks within the same Google Cloud project, deploy a Virtual Machine (VM) instance in each network, and establish bidirectional VPC Network Peering to enable private internal IP connectivity between them.

Learning Objectives

- Configure custom VPC networks with non-overlapping IP subnets.
- Deploy Compute Engine VM instances into distinct VPC networks.
- Configure VPC Network Peering connections from both directions.
- Set up firewall rules to permit ICMP traffic across peered VPCs.
- Verify internal connectivity between VMs in separate VPC networks.

Configuration Reference Summary

Resource Type	Resource Name	Region / Zone	Network / Subnet	IP Range / Settings
VPC Network A	`vpc-network-a`	`us-central1`	`subnet-a`	`10.1.0.0/24`
VM Instance A	`vm-a`	`us-central1-a`	`vpc-network-a` / `subnet-a`	Machine: `e2-micro`
VPC Network B	`vpc-network-b`	`us-central1`	`subnet-b`	`10.2.0.0/24`
VM Instance B	`vm-b`	`us-central1-a`	`vpc-network-b` / `subnet-b`	Machine: `e2-micro`
Peering A to B	`peering-a-to-b`	Global	`vpc-network-a` → `vpc-network-b`	Bidirectional (Part 1)
Peering B to A	`peering-b-to-a`	Global	`vpc-network-b` → `vpc-network-a`	Bidirectional (Part 2)
Firewall Rule A	`allow-icmp-vpc-a`	Global	`vpc-network-a`	Source: `10.0.0.0/8`, Protocol: `icmp`
Firewall Rule B	`allow-icmp-vpc-b`	Global	`vpc-network-b`	Source: `10.0.0.0/8`, Protocol: `icmp`

Implementation Steps

Task 1: Create VPC Network A and Deploy VM A

1. In the Google Cloud Console, navigate to **VPC network** > **VPC networks**.
2. Click **Create VPC network**.
3. Configure the parameters:
    - **Name:** `vpc-network-a`
    - **Subnet creation mode:** Custom
    - **Subnet name:** `subnet-a`
    - **Region:** `us-central1`
    - **IP address range:** `10.1.0.0/24`
    - Click **Create**.
4. Navigate to **Compute Engine** > **VM instances** > **Create Instance**.
5. Configure properties:
    - **Name:** `vm-a`
    - **Region:** `us-central1`
    - **Zone:** `us-central1-a`
    - **Machine Type:** `e2-micro`
6. Under **Networking options**, set:
    - **Network:** `vpc-network-a`
    - **Subnetwork:** `subnet-a`
    - Click **Create**.

Task 2: Create VPC Network B and Deploy VM B

1. Navigate back to **VPC network** > **VPC networks** > **Create VPC network**.
2. Configure the parameters:
    - **Name:** `vpc-network-b`
    - **Subnet creation mode:** Custom
    - **Subnet name:** `subnet-b`
    - **Region:** `us-central1`
    - **IP address range:** `10.2.0.0/24`
    - Click **Create**.
3. Navigate to **Compute Engine** > **VM instances** > **Create Instance**.
4. Configure properties:
    - **Name:** `vm-b`
    - **Region:** `us-central1`
    - **Zone:** `us-central1-a`
    - **Machine Type:** `e2-micro`
5. Under **Networking options**, set:
    - **Network:** `vpc-network-b`

- **Subnetwork:** `subnet-b`
- Click **Create**.

Task 3: Establish Bidirectional VPC Network Peering

1. Navigate to **VPC network** > **VPC network peering** in the Cloud Console.
2. Click **Create connection** > **Continue**.
3. Create **Peering Connection 1 (A to B)**:
   - **Name:** `peering-a-to-b`
   - **Your VPC network:** `vpc-network-a`
   - **Peered VPC network:** In this project → `vpc-network-b`
   - Click **Create**.
4. Create **Peering Connection 2 (B to A)**:
   - **Name:** `peering-b-to-a`
   - **Your VPC network:** `vpc-network-b`
   - **Peered VPC network:** In this project → `vpc-network-a`
   - Click **Create**.
5. Confirm that both peering connections show a **Status** of `Active`.

Task 4: Configure Firewall Rules for ICMP Connectivity

1. Navigate to **VPC network** > **Firewall** > **Create firewall rule**.
2. Configure Firewall Rule for Network A:
   - **Name:** `allow-icmp-vpc-a`
   - **Network:** `vpc-network-a`
   - **Targets:** All instances in the network
   - **Source IPv4 ranges:** `10.0.0.0/8`
   - **Specified protocols and ports:** Check **Other protocols** and type `icmp`.
   - Click **Create**.
3. Configure Firewall Rule for Network B:
   - **Name:** `allow-icmp-vpc-b`
   - **Network:** `vpc-network-b`
   - **Targets:** All instances in the network
   - **Source IPv4 ranges:** `10.0.0.0/8`
   - **Specified protocols and ports:** Check **Other protocols** and type `icmp`.
   - Click **Create**.

Task 5: Test and Verify Internal Connectivity

1. Navigate to **Compute Engine** > **VM instances**.
2. Note down the internal IP address of `vm-b` (e.g., `10.2.0.2`).
3. Click **SSH** next to `vm-a` to open a terminal session.
4. Test reachability to `vm-b` over internal IP:

   ```
   ping -c 4 10.2.0.2
   ```
5. Verify that all packets are successfully transmitted with 0% packet loss.

Exercise Cleanup

To avoid incurring unnecessary charges to your Google Cloud account, clean up the resources created during this exercise:

1. Delete VM instances `vm-a` and `vm-b` under **Compute Engine** > **VM instances**.
2. Delete peering connections `peering-a-to-b` and `peering-b-to-a` under **VPC network** > **VPC network peering**.
3. Delete firewall rules `allow-icmp-vpc-a` and `allow-icmp-vpc-b` under **VPC network** > **Firewall**.
4. Delete VPC networks `vpc-network-a` and `vpc-network-b` under **VPC network** > **VPC networks**.